

From Spare Compute to Cumulative Mathematics

An open-source strategy for agent-assisted research on hard problems

WILL COOK*

September 2026

ABSTRACT

This paper describes an open-source strategy for agent-assisted mathematical research. The starting point is a practical asymmetry. Building a durable research environment requires navigation, experiment records, formal proof checking, claim boundaries, review tools, and attribution machinery. Once that fixed cost has been paid, another person should not have to rebuild it in order to contribute a mathematical idea or a night of spare compute.

The proposed research commons accepts four independently useful inputs: compute, mathematical direction, infrastructure improvements, and review. Its outputs are not limited to solved problems. Proofs, counterexamples, finite computations, corrected statements, failed mechanisms, formal no-go theorems, and better research tools all reduce the cost or improve the direction of later work. Accepted mathematical returns enlarge a versioned problem corpus; accepted architectural returns improve the process that turns future compute and insight into reviewable mathematics. Provenance and role-specific credit remain attached when a contribution is assimilated.

The implemented case is a public Lean repository organised around eight Erdős problem programmes. Each has a short paper, longer research record and Lean source; a separate cross-problem paper joins the mathematics. The problems were chosen as difficult research worlds, not because the present system was expected to solve them. A fresh clone also contains explicit open boundaries, corpus queries, validation programs and return paths. The degree-seven counterexample due to the [erdosproblems.com](#) contributor [ani](#), formalised in Lean here for one degree-seven polynomial, gives a one-dimensional Hausdorff-measure bound and refutes the exact Formal Conjectures path-image-length statement of Problem 1041. The separate total-variation bound is also checked. The reported family is not formalised, and correspondence with the 1958 curve-length wording remains unreviewed. The other seven targets remain unresolved here. In the 257 lane, the repository proves all-base irrationality for full support, infinite reciprocal-summable supports, and supports meeting a base-two finite-prime weighted condition. The arbitrary infinite-support question remains open. A public architecture return has passed focused checks and entered the accepted contribution record with role-specific credit. Its reproduction was maintainer-operated by the same contributor; it does not show that an independent outsider can use the route. The repository thus demonstrates a contribution mechanism, while its benefit to unfamiliar readers and the effect of additional compute remain untested.

The paper gives a production model, contribution protocol, trust and security boundary, attribution policy, comparison with volunteer computing, Polymath-style collaboration, formalisation projects, the Prove2Me platform, the Aletheia case study, and multi-agent proof systems, and an evaluation agenda. Mathematics is the initial domain because its objects are digital and formal claims can be checked incrementally. Any later extension to experimental science would require domain scientists, physical laboratories, safety governance, and new evidential authorities.

**Authorship and AI use.* Every word of this manuscript was generated by agents based on large language models operating within Will Cook's private research system for artificial intelligence. Cook set the objectives and acceptance criteria and is responsible for its contents. The agents are production tools, not authors. This production disclosure supplies no independent verification; Cook did not independently verify every mathematical claim.

What is implemented

Contribution. A public contribution protocol accepts compute, mathematical direction, infrastructure, and review as independently useful inputs, while preserving evidence boundaries and role-specific credit. **Implementation.** One fresh clone contains eight problem worlds, formal source, explicit open obligations, validation programs, and return paths. One architecture return was accepted on 23 September after focused same-contributor reproduction. **Limit.** Independent first use and any gain over good static materials have not been measured.

1 The strategy

This project began with one undergraduate working without an institutional research team or a dedicated compute allocation. That fact does not validate the mathematics or the architecture. It makes the strategic question concrete: can the fixed cost of research infrastructure be paid once, published, and then shared by people who bring different scarce inputs?

The intended answer is an open research commons. A mathematician may supply a theorem, counterexample, reference, or correction. A Lean contributor may formalise or audit one statement. A person with spare compute may run an agent against a bounded public frontier; at present that means choosing a bounded task, operating it through a runner of their own, and returning evidence, not installing a turnkey client. An infrastructure contributor may improve navigation, experiments, validation, reproducibility, or the public contribution path. A reviewer may reconcile a formal statement with its intended meaning or decide that a returned result does not survive scrutiny. These are different contributions. None should have to masquerade as a solution to receive credit.

The project is AI-native in a narrow, declared sense. Contributors may use language models and agent harnesses throughout the work, provided that their use is disclosed. Model assistance is neither a defect nor evidence of mathematical value. What enters the accepted record is the attributable delta: a mathematical idea, a question or direction that produces useful mathematics, a proof or counterexample, a review, or an architectural change that makes later research more effective. Each still needs the evidence and review appropriate to its claim.

The repository is deliberately centred on hard problems that current systems cannot simply dispatch in one prompt. The purpose is to expose the complete research process: selecting a useful subproblem, reading prior work, testing conjectures, recording failed routes, formalising stable steps, explaining the result, and preserving the exact statement that remains open. A solution is one possible output of this process. The cumulative record is the output that can be produced on every serious run.

Each problem is therefore treated as a continuous goal rather than a sequence of blank-slate prompts. The endpoint persists while the frontier changes. A new agent run starts from the accumulated theorems, computations, counterexamples, failed mechanisms, citations, and open obligations; chooses a direct mathematical transition worth attempting; and returns an inspectable delta plus the boundary that survived. It may hand back a theorem, a narrower reduction, a killed route, a source correction, or a better next question. When the endpoint remains open, that is not a failed run. Losing the state and paying to rediscover it would be.

A listed endpoint is one way to begin. The other is to read the corpus as a single body of mathematics and decide what is worth developing. Eight programmes written side by side expose methods, obstructions and examples that no one programme sets out to study, and a good question is scarcer than the compute to attack it. The commons therefore supports a run whose question, subject and eventual form emerge during the work. The constraint that remains is source fidelity. The recorded sources fix what may be claimed about existing results, and they place no limit on the question asked. Such a run returns the same kind of object as any other:

a stated idea, its evidence, the prior work found, its limits, and the next question it opens. The public repository carries one such investigation in full, including the step at which it corrected its own first reading of its data ([choices](#), [contraction](#) and [rational membership](#)).

A second cross-problem investigation gives a small example of why returned work needs an exact limit. The rational number $189/388$ survives the first sixteen steps of a Mersenne greedy test and is rejected at step seventeen by exact rational arithmetic. This refutes a proposed extrapolation from a fixed-depth counting theorem; it does not make finite survival a certificate of membership. The [reproduction program](#) and the [synthesis paper](#) give the arithmetic and the surviving theorem. It is a contribution even though it closes no listed endpoint: the next run no longer starts from the false stopping rule. The [exact-rational probe](#) varies host and finite bounds; non-exclusion at a tested depth is not membership. The record lists five next tests and where to report results.

Two coupled goals, not one endless agent

The most useful public deployment has two persistent roles. A *discovery goal* stays close to one mathematical frontier: it reads the corpus, compares attacks, uses computation to discriminate them, attempts ordinary and formal proofs, and returns the smallest stable theorem, counterexample, no-go, computation, or corrected boundary. A *stewardship goal* stays close to the corpus as a whole: it compares the new object with existing results, decides which declarations form one mathematical family, reconciles the paper and assurance surfaces, and returns the strongest next question to the discovery goal. The corresponding systems contract is stated in the [coupled-goals section of the systems paper](#).

The two roles can be run by different people, agent harnesses, or machines. A compute contributor may operate only the discovery side. A mathematician may supply a direction, assess whether a result is genuinely nontrivial, or repair the statement and exposition without paying for a long model run. An expositor may improve the paper projection; a Lean contributor may strengthen the formal interface; an agent builder may make the loop cheaper. All of these acts remain attributable contributions to one evolving mathematical record.

The stewardship role is not a ceremonial reviewer added after the work. Its appraisal changes where future effort goes. It may discover that a new “theorem” is only a reformulation, that five declarations are one coherent result family, that a short no-go eliminates an expensive research direction, or that a paper still leads with a weaker theorem. It then updates four separate outputs: proof and evidence status, mathematical appraisal, paper prominence, and the next allocation of compute or expert attention. These outputs may influence one another, but they are not one score and none can promote an unproved claim.

Conditional proof cost offers one way to propose that allocation: Patel et al. [13] rank statements by proof cost relative to statement length and compare this with reuse. The systems paper tests such a proxy on this repository before using it for triage. Shared factors in the scores, generated certificates and duplicated interfaces can all affect a ranking; the stewardship procedure asks which concrete consumer, containment check or failed route deserves investigation. Whether this allocation improves later research requires a prospective comparison with simpler baselines.

The coupling is event-driven. A stable mathematical delta wakes the steward; a changed appraisal, missing consumer, or sharper open boundary can wake the miner. An unchanged repository should consume no agent turn merely to report that it is unchanged. This makes continuous work a sequence of inspectable state transitions rather than an expensive synonym for leaving a chat window open.

The public [run-coupled-research-goals skill](#) makes this control shape executable in a cold clone. It invokes the existing mining and consequence-propagation jobs, preserves a shared source pin, and passes committed objects and receipts rather than conversational claims of

progress. One agent may alternate between the roles, or different people, models, subscriptions, and machines may supply them. The architecture requires distinct decisions, not an unnecessarily grand collection of laptops.

The public Lean clone owns the mathematics and return path. The private workbench explains production, not proof authority, and contributors need no access to it. Any runner or no model at all can use the public evidence and review path. The [public website](#) is a rendered reading route. Its canonical generator and editorial configuration remain private; the Lean clone is the inspectable source.

The [systems paper](#) gives the complete claim-transition lifecycle. The [cold-clone paper](#) tests how a new agent reaches the public mathematics and the validation boundary. This paper owns the participation, credit, and growth strategy; the three papers therefore describe different parts of one prototype rather than competing accounts of it.

What one clone lets a contributor do

A contributor does not need to understand the whole repository before doing useful work. From one clone, a person or agent can choose one of six first actions: read the corpus as one body of mathematics and decide what is worth developing; mine a bounded problem route; contribute mathematical direction without paying for the compute that follows it; formalise or review one claim; repair the research machinery; or propose another sourced problem world. The first action has no target at the start. Finding a question that deserves attention is part of the research, and a well-posed new question, a general theorem behind a recorded example, or a clearer account of an existing proof is a contribution in the same sense as a proof. The [explore-the-corpus skill](#) owns that route and hands an emerged question to the workflows below. The clone-local [explain-public-system skill](#) lets an agent read the public corpus and companion papers on the newcomer's behalf, explain the claim boundaries at the requested level, and point back to exact evidence. The reader can therefore begin with the ordinary request “explain this repository to me” rather than first mastering its file layout. The other skills select and run a frontier, coordinate the coupled goals, install the same workflows in a compatible agent harness, and describe the present multi-stage process for adding a problem.

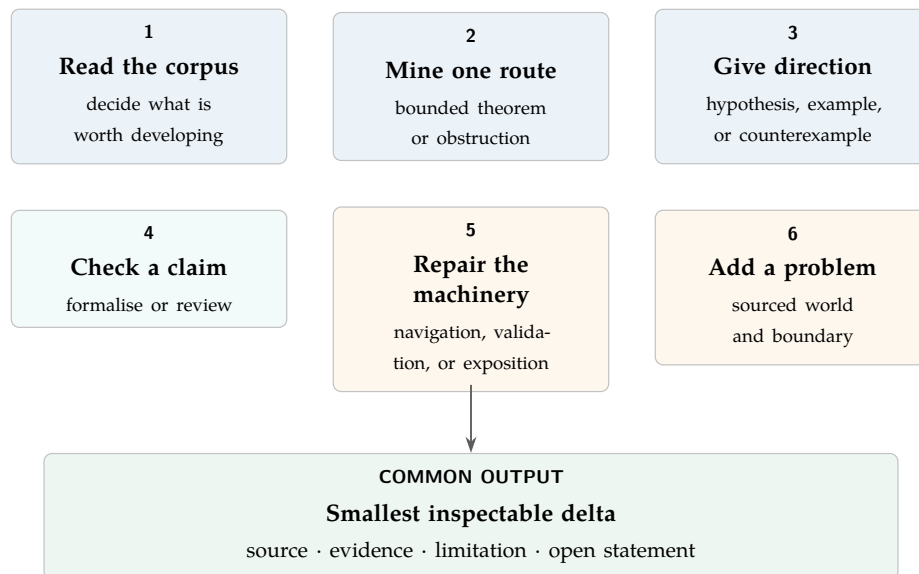


Figure 1: Six useful first actions. None requires an endpoint solution, and none receives a stronger status than the evidence returned with it.

2 A production model for mathematical progress

Candidate production depends jointly on compute, model capability, human direction, infrastructure, problem choice, and accumulated knowledge. Mathematical progress additionally depends on validation and scarce human review. The notation below makes those dependencies explicit. Let

$$C_t, A_t, H_t, I_t, P, K_t, V_t, R_t$$

denote, at time t , available compute, agent capability, contributed human mathematical direction, infrastructure quality, problem selection, accumulated research knowledge, validation capacity, and review capacity. The rate of candidate generation may be written schematically as

$$G_t = G(C_t, A_t, H_t, I_t, P, K_t). \quad (2.1)$$

The rate of *reviewable mathematical progress* is a different quantity:

$$Y_t = Q(G_t, V_t, R_t), \quad (2.2)$$

where Q includes formal checking, statement reconciliation, attribution, and editorial selection. Equations (2.1) and (2.2) are a conceptual production model, not a fitted empirical law. They record complementarity and bottlenecks. More compute can generate more attempts without increasing the rate at which strong results are validated or understood. A better model can still revisit a closed route if the corpus is difficult to navigate. Expert insight can change the search distribution without supplying any compute. Review capacity can be the binding constraint even when proof generation is cheap.

The open-source strategy proposes two recursive updates:

$$\begin{aligned} K_{t+1} &= K_t + \Delta_t^{\text{theorem}} + \Delta_t^{\text{counterexample}} + \Delta_t^{\text{no-go}} + \Delta_t^{\text{computation}} + \Delta_t^{\text{correction}}, \\ I_{t+1} &= I_t + \Delta_t^{\text{navigation}} + \Delta_t^{\text{validation}} + \Delta_t^{\text{reproducibility}} + \Delta_t^{\text{governance}}. \end{aligned} \quad (2.3)$$

These Δ terms are *net changes to the currently endorsed record*, not nonnegative counts of retained files. A correction may withdraw an earlier claim, replace a proof or close an unproductive route, so K_{t+1} need not exceed K_t even though the provenance history is append-only. Reviewed, scoped returns may make later research less forgetful or infrastructure more reliable; the equations do not estimate either gain. A local failure can add a mathematical obstruction to the record or reveal a process defect whose repair improves infrastructure. It cannot change a conjecture's truth status merely because it changed the process.

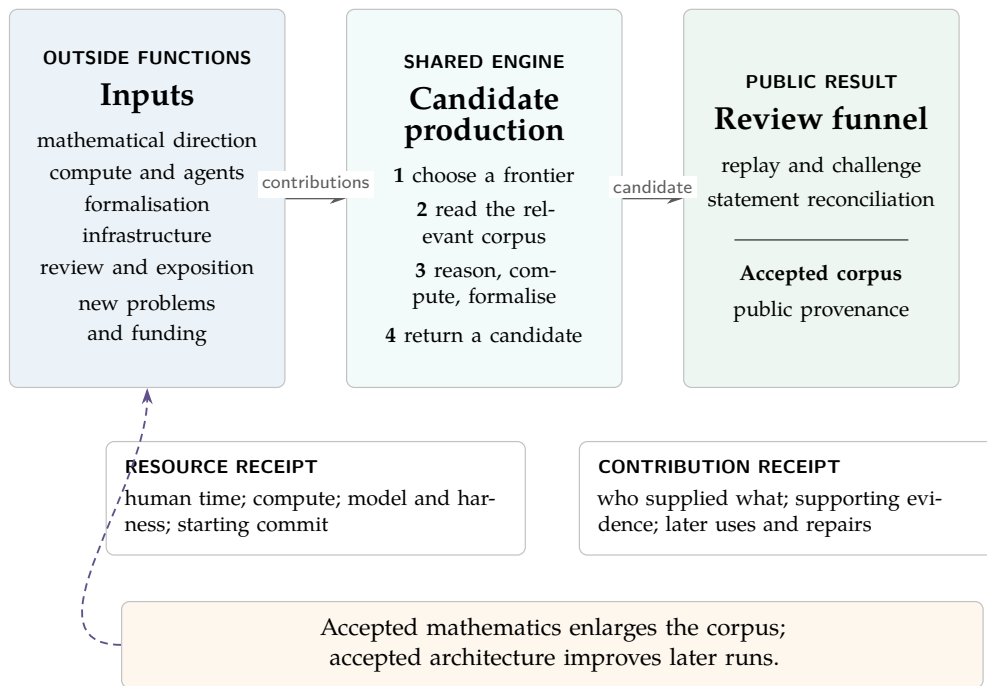


Figure 2: The conversion loop. The labels on the left are functions, not classes of people: one person may perform several, and several people may perform one. Search output crosses replay and review before it enters the accepted corpus. Mathematical returns enlarge the corpus; architecture returns alter the engine.

The model also gives a longitudinal use for the repository. A fixed, versioned problem world can be revisited as models, runners, and compute budgets change. Comparisons must disclose the starting commit, model, scaffolding, number of attempts, compute, and review procedure. Otherwise a success says little about which variable changed. A public failure record is equally important: capability claims are badly distorted when successes are announced and the number and cost of failed attempts are hidden [1].

3 Why begin with hard mathematics

Mathematics is an unusually clean first domain for this strategy. The objects, programs, papers, and formal statements can all travel in one clone. Many conjectures can be probed by exact computation. A proof assistant can check a stable formal step without waiting for an entire long argument to be written informally and retrofitted later. This does not make mathematical meaning automatic: Lean checks the proposition in the source, not whether it is the proposition the project intended to study.

Computation has a second role. It can provide an artificial form of local intuition to a system that is better at writing and running code than at reproducing a mathematician’s tacit judgement. Exact experiments can expose counterexamples, compare representations, and locate a narrow regularity. Their result updates route selection and conjecture pressure. It does not change the evidence class of an unbounded statement. In Bayesian language, an experiment can update which route deserves attention; it cannot assign a formal posterior probability to a theorem or replace proof.

The eight current problems were selected as demanding research environments, not as a claim that they were the eight most tractable or valuable open problems. Their papers expose materially different contribution points.

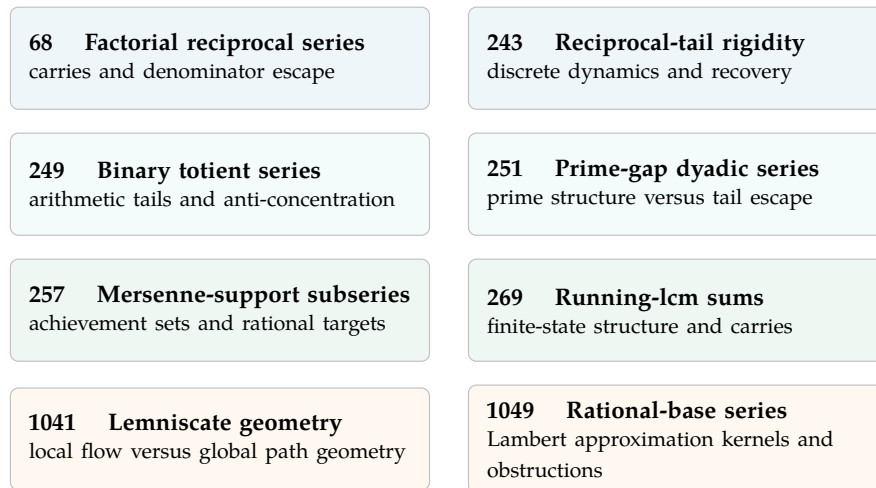


Figure 3: Eight distinct research environments. The results map and the problem papers state their exact frontiers; this strategy paper does not reproduce the specialist vocabulary needed to attack them.

Every row admits more than theorem proving. A mathematician may sharpen a hypothesis or supply a counterexample. A programmer may run a discriminating exact calculation. A formaliser may turn a paper deduction into a checked declaration or find that the two statements disagree. A literature reader may locate a theorem that closes or invalidates a route. The contribution is the exact returned object and its limitation, not the prestige of the problem number.

The different evidence classes matter in this portfolio. In the 257 lane, a Lean-checked finite-prime weighted condition on an infinite exponent set A makes $\sum_{a \in B} (b^a - 1)^{-1}$ irrational for every integer $b \geq 2$ and every infinite $B \subseteq A$. The paper constructs such an A with divergent $\sum_{a \in A} 1/a$; arbitrary infinite supports remain open. Erdős had stated the narrower reciprocal-summable extension. The 249 lane checks certificates at every lcm-diagonal scale through 82; the unbounded supply needed for irrationality has not been proved. In the 1041 lane, the explicit degree-seven example due to an refutes the Formal Conjectures Hausdorff path-image claim, while its relation to the historical curve-length question awaits independent human review. A participant can advance any of these worlds by supplying the next mathematical fact, a correction, an exact computation, or a review that keeps those boundaries visible.

4 The public research object

The unit of participation is a *problem world*. A problem world includes the endpoint question, current public status, principal theorems, formal source, experiments, known counterexamples, failed mechanisms, source literature, and exact open obligations. A bounded task is a neighbourhood in that world, not an invitation to read the repository indiscriminately.

The research was initially developed inside a private workbench. The public release carries both the mathematical corpus and the workflows needed to inspect, validate, and return a contribution. A contributor uses the public checkout as the working environment; access to the private workbench is not part of the contribution protocol, and the workbench grants no proof or publication authority to a returned result.

The public layers carry different kinds of evidence. An ordinary proof, a cited theorem, a finite experiment and a Lean proof can each support a candidate on its stated domain. Review asks what that evidence establishes; formal statement alignment applies when a Lean claim is made. Uptake by other researchers is a separate observation.

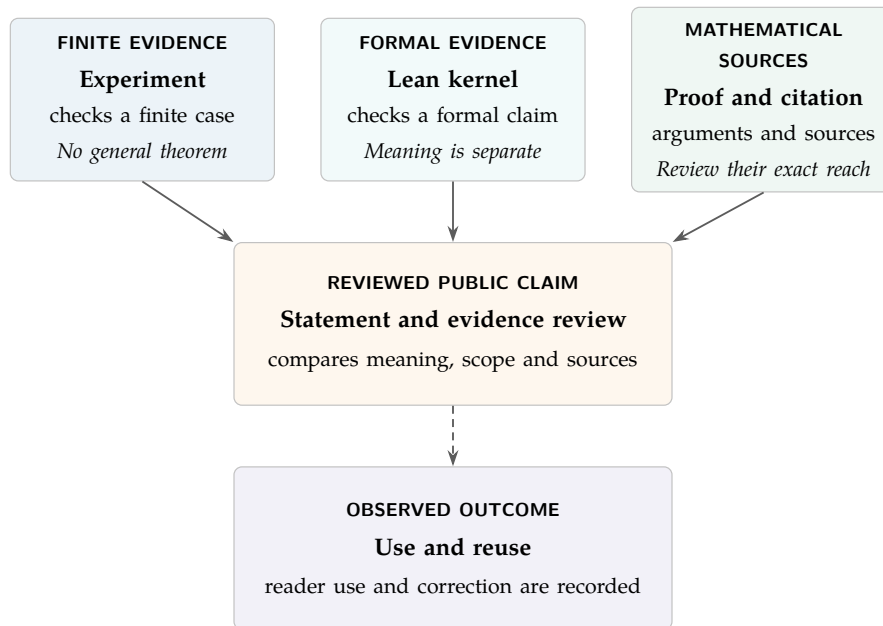


Figure 4: Evidence routes keep their own scope. Formal claims require a meaning check; ordinary proofs and citations do not require Lean. Outside use is recorded separately.

The project is a composition layer around substantial prior work, not a replacement for it. Erdős Problems supplies questions, sources, status, and a problem community [7]. Lean supplies the formal language and kernel [8]. DeepMind’s formal-conjectures supplies a public one-problem-file model with metadata, snapshots, issues, and pull requests [9]. *Comparator*, the Lean FRO’s statement checker, compares selected exact statements with their Lean implementations and permitted axioms. A repository-local review ranks candidate results and checks whether the selected family is ready for possible submission to Palomar [10]. That readiness is not an external Palomar decision and does not establish novelty or registration; exact-head *Comparator* replay is reported separately. Polymath supplies a social precedent for publishing small, tentative, and negative contributions [4]. BOINC and GIMPS supply the volunteer-compute precedent and make visible why executable work, independent checks, and legible credit matter [2, 3]. The present repository wraps these practices into a route that a newcomer can enter without first rebuilding each component.

The same underlying substrate can support several reader projections: a short public primer, a specialist paper, a detailed proof account, Lean declarations, an agent explanation, and machine-readable queries. Each projection must link back to its source claims and evidence. A shorter or friendlier account does not acquire permission to strengthen them.

An external runner begins at [the compact agent entry](#). It can inspect the problem frontiers, choose a bounded question, read the relevant paper and source, run experiments or edit formal code, validate the result, and prepare a return. None of this requires the private workbench. A reader with no clone starts from the [reading edition](#). Its two committed files contain an introduction and research instruction, the cross-problem paper, and either short-paper openings or all eight short papers in full; an edition with all 17 mathematics papers can be built on demand. The builder’s `--check` detects stale outputs against the paper sources. File sizes are printed so a reader can choose what a model can load. A contributor is free to replace the agent, the scheduler, or the entire search policy while retaining the public evidence boundary.

5 The contribution protocol

Patches return by pull request. An idea, reference, correction, or result needs no clone or Lean: a plain-language research-progress issue or email can start review. Infrastructure ideas use the architecture-proposal path. Adopted ideas and later proofs or implementations receive distinct credit.

The clone-local submit-pull-request skill makes this sequence agent-readable. It separates mathematically or architecturally coherent changes into reviewable commits when they can genuinely be separated, runs the narrow checks for each change, and prepares the repository's pull-request template. Commit boundaries follow the meaning of a change rather than a rule of one file or one diff type per commit. Pushing to a fork and opening the pull request are external actions and occur only after the contributor authorises them.

Upstream work may continue while the contributor's branch remains open. The recorded starting commit supplies the common ancestor from which Git recovers the original delta; the contributor does not need a copy of the maintainer's private or current working tree. Review therefore has two passes. First, reproduce the change in its original public context. Second, reconcile the accepted substance with current main and rerun the present checks. A material conflict resolution is a new, separately credited integration change. It does not erase or retrospectively rewrite the original contribution.

For an idea-only submission, maintainers or later patch contributors perform the validation steps in the nine-step adoption path below.

1. **Anchor.** For a patch, record the exact public commit and environment. For an idea, cite the problem, paper, or source if known.
2. **Orient.** Read the claim boundary and the paper governing the selected problem or architecture area.
3. **Select or discover.** Name one bounded frontier, expected evidence, and a stop condition. Working on all problems is allowed, but each returned result remains independently scoped. A run may instead begin from the corpus with no frontier named. The question is then an output of the reading, and it is stated with the same three items as soon as it has emerged.
4. **Work.** Use any mixture of reasoning, source research, computation, Lean, and human mathematical judgement. Preserve useful negative results instead of rewriting the run as a linear success.
5. **Validate.** Run the checker appropriate to the claim. A computation receives a computation receipt; a formal theorem receives a Lean receipt; neither receives semantic or publication authority from that fact alone.
6. **Propagate.** Enumerate the plausible Lean, claim, paper, computation, route, validation, and contributor consumers. Update each one, verify it unchanged, defer it with a re-entry condition, or explain why it is not a consequence. Repeat this pass after old-branch work is reconciled with current main.
7. **Return.** Submit the idea or artifact, evidence, limits, collaborators, and preferred credit; add commit and tool details when relevant.
8. **Review and adopt.** Reviewers reproduce the evidence, reconcile formal and informal statements, decide scope, and either accept, request revision, reject, or preserve the return as a scoped negative result.

9. **Publish and credit.** An accepted generation is linked to the paths and commit that assimilated it. Later corrections append lineage; they do not erase the earlier contributor.

The accessibility claim is role-bounded. A non-specialist compute contributor is not asked to decide what constitutes a proof or to invent a mathematically meaningful objective unaided. The default work unit is a packet prepared from an authored mathematical route: an exact open statement, a relevant source neighbourhood, a permitted experiment or proof obligation, an expected evidence class, and a stopping condition. Qualified mathematical review should precede large-scale distribution; it is not claimed for every present route. The contributor and agent execute the packet and return evidence. Feng et al. report a case in which stating the admissibility rules of the target, after a first attempt had cited results that likely do not exist, cut the inference needed for a correct solution sixteenfold on their scale [16]. A task packet is that rule statement written down once, before the compute is spent. Lean checks a formal proposition. A qualified reviewer checks whether that proposition says what was intended and whether the return is relevant. The return remains a candidate until the applicable gates have been crossed.

The mathematician who supplies an input and the mathematician who verifies a return are also distinct roles. The first may contribute a conjectural mechanism, a counterexample pattern, or simply a well-posed question that causes the system to produce useful mathematics. The second inspects the resulting statement, argument, prior art, and significance. One person may perform both roles, but the provenance record must not let an input certify its own output merely because it came from a mathematician.

This protocol supports two first-class tracks. The *mathematics track* accepts proofs, reductions, computations, counterexamples, corrections, formalisation, and reproducible failed routes. The *architecture track* accepts improvements to agent workflow, navigation, validation, reproducibility, public experience, governance, and tooling. An architecture return never needs a fictitious problem number, and it cannot request promotion of a mathematical claim. The same holds inside the mathematics track. A return whose subject is wider than one problem, such as a general theorem, a mechanism shared by several programmes, or a new question with evidence for its value, names that subject in words and lists the related problems. The list may be empty. The credit record names the subject and related problems. The public *architecture route* accepts an observable failure and stop condition, such as repairing a missing cold-clone validation route and checking its discovery. An adopted design is credited without promoting a mathematical claim.

The public *accepted-contribution record* gives one architecture example. An OpenAI Codex return repaired the ordinary-language route into the research-return workflow and added a checked-positive architecture packaging case. Focused source checks were reproduced, the maintainer recorded adoption and the receipt keeps the software and validation credit. The contributor and maintainer-operated reviewer were the same agent; neither the receipt nor this paper claims an independent outside clone replay. A mathematical idea, an ordinary proof and a Lean proof would require their own corresponding review.

The distinction between a return and an adopted result is load-bearing. A pull request is a proposed delta. A review act records a decision. A release generation is an immutable public state. Keeping these separate permits credit for a useful counterexample, correction, or design even when no theorem is merged.

6 From an agent claim to mathematical attention

An agent saying “solution found” is an event in a run log, not a project claim. An informal derivation may contain an unnoticed gap. A Lean theorem may be completely kernel-correct while formalising the wrong statement, assuming an inadmissible axiom, or proving a result too weak to settle the problem. Formal verification therefore strengthens one layer of evidence;

it does not collapse correctness, intended meaning, novelty, significance, and community acceptance into a single status.

Feng et al. measured this gap on the Erdős problems themselves [15]. Their agent worked the 700 problems then marked open in Bloom’s database; human graders assessed 200 candidate solutions, found 63 technically correct and 13 meaningfully correct, and traced the fifty vacuous solutions to statements that had been misread or mistranscribed. The lengthiest step of their study was deciding whether a correct solution was new and whether it addressed the intended problem. Their pipeline placed an automated natural-language verifier before scarce human experts, and the funnel below has the same shape. Their companion report states the premise of this section outright: many open questions lie within reach of existing techniques and stay open for want of time and attention from the right experts, and machine effort is bottlenecked by different factors [16].

The project uses a deliberately asymmetric attention funnel:

1. **Candidate.** A person or agent returns a scoped argument, computation, formal declaration, counterexample, or no-go.
2. **Replay.** Automated checks reproduce the code and attach the correct evidence class without promoting its claim.
3. **Internal challenge.** Independent agents or contributors try to break the statement, locate stronger prior art, vary the experiment, and reconcile formal and informal formulations.
4. **Maintainer triage.** A human checks whether the bundle is coherent enough, relevant enough, and sufficiently de-risked to justify scarce specialist attention.
5. **External record.** A mature Lean result may be packaged through Comparator and submitted to Palomar; a mathematical account relevant to a listed problem may be placed before the Erdős Problems community and the appropriate research audience.
6. **Acceptance.** Mathematicians inspect, explain, criticise, reuse, publish, or reject the work over time. Broad mathematical acceptance is exogenous to this repository and cannot be granted by its maintainer.

Step three follows Henkel, who proposed a different model for verification than for generation on the evidence that each model is worst at grading its own proofs [17, 18].

Palomar is valuable precisely because its claim is narrower than peer review. It mechanically verifies a pinned challenge–solution interface using Comparator and kernel checking, records structured disclosure, and applies a documented automated editorial filter. It explicitly does not endorse the result, establish novelty, or provide human expert review [10]. Likewise, the Erdős Problems forum asks long proofs or partial proofs to be linked as external documents and filters implausibly incomplete claims; it is a route to relevant attention, not a substitute for that community’s judgement [7].

Multiple contributors can improve triage, but headcount is not mathematical evidence. What raises a bundle’s priority is costly, legible corroboration: independent replay, adversarial review, distinct contributors repairing different failure modes, a stable formal interface, and an exposition that a specialist can audit. This social proof should allocate attention, never inflate the theorem status. It lets a contributor without institutional connections accumulate a public chain of reasons for an expert to look, without requiring the expert to sift every raw agent transcript.

7 Progress, provenance, and credit

The project recognises progress in evidence classes rather than by counting commits or theorems. A return may contain:

- an unconditional theorem or a formal proof of an existing statement;
- a counterexample or corrected hypothesis;
- a no-go theorem excluding a defined strategy class;
- a bounded computation with code, range, and interpretation;
- a conditional reduction with its antecedent still visible;
- a reproducible failed route that prevents duplicated search;
- a literature correction or exact source locator;
- an architecture proposal or validated infrastructure change; or
- exposition or review that changes what another person can understand and check.

The need to retain such work appears in another study. Feng et al. found that literature identification was the lengthiest step of their Erdős study and that misread statements produced most of their vacuous solutions; they proposed revised statements as a deliverable [15]. This observation does not measure the payoff of this project’s credit scheme.

These items should not be placed on one scalar leaderboard. Compute volume, mathematical depth, software quality, correction work, exposition, and review are not commensurable. Feng et al. propose a public two-axis classification for AI-assisted results, level of autonomy (primarily human, human–AI collaboration, essentially autonomous) against level of mathematical significance (negligible novelty to landmark breakthrough), and hold that only domain experts can place a result on the second axis [16]. The role ledger here is compatible with that scheme and finer-grained on the first axis: it records which role each contributor played and which evidence class each artefact carries, from which an autonomy level can be read off. The significance axis remains an expert judgement that the ledger does not attempt to fill. The same report calls for transparent standards for how AI-generated mathematics is documented, evaluated, and communicated, and offers its taxonomy and cards as a starting point; the protocol, evidence classes, and receipts here are one such standard, offered in the same spirit. A public recognition view can instead expose the contributor, accepted artefacts, evidence classes, problem or architecture area, and contribution roles. The current receipt schema supports the fourteen CRediT roles as an optional vocabulary, including conceptualisation, methodology, software, validation, investigation, and writing [20]. CRediT describes work; it does not decide authorship.

The default promise is simple: the project does not demand ownership of a contributor’s underlying idea in exchange for adoption. If a mathematical or architectural contribution is assimilated, the public lineage should continue to name who originated it and what survived. Substantial authorship and formal publication decisions remain separate judgements. Feng et al. hold that mathematics papers should always be authored by humans because authorship carries accountability for validity and attribution [15]; this project applies the same rule, and its manuscripts name one human author with the agents disclosed as production tools. A role ledger is more honest than either a winner-takes-all solver credit or a pseudonymous collective record when the project can preserve exact provenance.

The same rule applies when a local result is carried into a broader library. The ledger should distinguish the person or run that produced the local observation from the expert who recognised its natural generality, reconciled it with prior art, designed the library interface, rewrote or formalised the proof, and stewarded review. If that expert turns a problem-specific result into a Mathlib-quality contribution, the project does not demand ownership of the resulting pull request or paper. It asks for a reasonable provenance link back to the route from which the idea

arose, while crediting the expert fully for the mathematical judgement and upstream work they actually performed. Provenance is not a device for seizing authorship in arrears.

This policy is deliberately generous because attribution is part of the growth strategy. A stranger needs a reason to spend thought or compute on a project whose open problems remain unsolved. Durable, inspectable credit for partial and negative work is one such reason. It is also technically useful: provenance makes later correction, reproduction, and literature attribution possible.

For a contributor outside a university, research institute, or AI laboratory, the accepted artifact is also something concrete they can point to. The receipt identifies the idea or implementation they supplied, the public paths that carry it, the evidence reviewers checked, the roles they performed, and the boundary that remained. This is not a degree, peer review, employment recommendation, or guarantee that another institution will value the work. It is a durable public account of an actual mathematical or architectural contribution, rather than a claim stranded in a private conversation or an unattributed model transcript.

Record now, trace consequences later

The first ledger should remain descriptive. A contribution receipt records who supplied which object, from which public state, under which role, with which evidence and limitation. It need not guess the contribution's eventual importance. Later accepted work can add reviewed lineage edges such as *uses*, *enables*, *repairs*, *formalises*, *refutes*, or *explains*. A consequence view can then show which later results, tools, or expositions depend on an earlier contribution. Prove2Me already implements the simplest such edge as a credit rule: later proofs that import a statement credit it, to its proposer in the paper and its prover in the live FAQ [14].

This later view begins with an immediate obligation. Before a result is returned, the clone-local propagate-research-consequences skill gives every plausible downstream consumer a disposition. The later graph may reveal further consequences as the corpus grows, but it does not excuse leaving known current consumers stale.

This is more informative than awarding points at intake, but it is not an objective measure of impact. Time does not establish causation, and a graph edge is an authored claim that may need correction. The view should therefore remain inspectable, non-scalar, and corrigible. A small observation may later have many consequences; a large compute run may have none beyond its negative result. Both retain their original receipt.

Any financial reward scheme would require a separate prospective policy for eligibility, conflicts, funding, tax, dispute, and the treatment of old receipts. Existing attribution must not silently become a financial contract. The graph can record that an idea mattered; it cannot, at present, send an invoice. Pure mining runs can still disclose the model, provider, harness, compute donor, starting commit, and any human direction as distinct roles.

8 Distributed compute without distributed authority

Volunteer compute has an established precedent. BOINC packages scientific jobs for heterogeneous consumer devices, and BOINC Central explicitly aims to make volunteer computing available to scientists without the resources to operate their own project [2]. GIMPS shows a mathematical version: volunteers run a common search program, independent machines confirm a prime, and discovery credit includes the compute donor, software authors, server operator, and wider volunteer effort [3].

Agent-assisted research is less uniform than either example. Most tasks are not interchangeable work units with a predetermined verifier. A runner may change code, propose a conjecture, or misread the problem. The public system therefore distributes *search* while keeping authority local to each evidence type. Mathematicians and formalisation contributors design or review

the routes; volunteers may execute them with Claude Code, Codex, Cursor, Antigravity, another open or proprietary harness, or no agent at all. The runner is replaceable. Prove2Me takes the same position for a hosted platform: it accepts any shell-capable agent and reports missions closed on ordinary consumer subscriptions [14]. The task packet, evidence boundary, and return record are the shared protocol. Returned code is untrusted until reviewed. Expensive or privileged continuous-integration jobs must not execute fork code with repository secrets. In particular, a GitHub `pull_request_target` workflow must not check out and execute an untrusted fork; GitHub documents that combination as a route to secret leakage and repository compromise [21]. Initial checks should run with read-only permissions, no secrets, bounded resources, and explicit artefact retention. Promotion to trusted infrastructure is a later review decision.

A useful compute return records at least the source commit, model and runner, tool disclosure, prompt or task packet, wall-clock time, human time, token or subscription use, and hardware budget where available, together with commands, changed paths, outputs, validation receipts, failures, and the stopping rule. These inputs should not be forced into one exchange rate: a mathematician's hour, an API bill, and a donated graphics processor are different resources. They can still be disclosed well enough to make a capability comparison interpretable and to reveal when hidden scaffolding paid most of the cost. Feng et al. propose a human-AI interaction card, a short record of each human query and each model return with the raw prompts and outputs published, as a transparency baseline for results where AI input was essential [16]. The return record here is that card with the evidence class and validation receipts attached.

The first public mode may remain deliberately simple: contributors clone the repository and run their own agents locally. A later volunteer-compute layer could distribute signed, immutable task packets and receive result bundles without granting write access. It should be built only after task identity, sandboxing, deduplication, resource budgets, result replay, and abuse handling have explicit owners.

9 Precedents and the missing composition

No part of this strategy is without precedent. The project deliberately packages practices developed by the Erdős Problems project, Lean and mathlib, Comparator and Palomar, DeepMind's formal-conjectures, Polymath, BOINC, GIMPS, the Prove2Me platform, and recent multi-agent formalisation systems. The intended contribution is an accessible composition: a person should be able to enter through one clone without first recreating those infrastructures.

Distributed compute. BOINC and GIMPS show that members of the public will donate hardware to a scientific objective when the client is easy to run, the work is visible, and credit is legible [2, 3]. Their tasks are much more mechanically uniform than open-ended proof research.

Distributed mathematical insight. Polymath projects invite participants at different mathematical levels to share small observations as they occur. Their rules explicitly welcome tentative and negative insights, provided they are made clear enough for others to absorb, and treat the project as collaboration rather than a race [4]. Polymath supplies a social protocol; it does not supply a formal proof, computation, and agent-return substrate for every comment.

Distributed formalisation. Mathlib uses ordinary fork-and-pull-request practice, human review, and source-level attribution. The Carleson formalisation used a public blueprint and many claimable lemma-sized tasks, with formalisation feeding corrections back into the informal plan [5, 6]. Google DeepMind's formal-conjectures repository similarly uses one-problem files, issue assignment, pull requests, metadata, and stable snapshots for a growing Lean benchmark [9]. These projects demonstrate that large checked mathematical objects can be made publicly divisible.

Multi-agent formal research. Agent Hunt studies bounties, locks, guarded ownership, and collaborative agents for autoformalisation [11]. Lean Atlas uses formal dependency information to reduce the declarations a person must inspect for semantic verification [12]. These mechanisms address search, coordination, and review focus. They do not by themselves provide a public commons in which mathematics and improvements to the research architecture enter one provenance-preserving adoption path.

Individual research practice. Henkel gives a framework for one mathematician working with publicly affordable models: five guiding principles, of which critical verification of every output is the second, and seven applications across the research lifecycle from ideation to writing [17]. He proposes cross-model checking on the evidence that each model grades its own proofs worst [18], and he treats acknowledgement of tool use as an emerging norm while recording that community standards for it remain to be established. His unit is one researcher at a chat window. The commons here supplies the shared record, evidence classes, and credit machinery that such a norm needs in order to hold across contributors.

Open agent platforms. Prove2Me is an open collaborative platform for formalisation in which anyone with a coding agent is a first-class contributor [14]. Its first arXiv version appeared on 28 August 2026, before the first public version of this paper, and it had first several of the ideas proposed here. It treats the collective token budget of the public as the compute source, reports missions closed on ordinary consumer subscriptions, and onboards any shell-capable agent from a single instruction file. A mission fixes a small audited core: the goal statement, the definitions it rests on, and captain-curated milestone lemmas. Agents introduce every other intermediate theorem freely. A proof-sketch is a sorry-free proof that imports statements not yet proved, so the target is verified once its imported children are, and proof search decomposes into atomised sub-problems that different agents can close. Every accepted statement enters a searchable reusable library, and later proofs that import a statement credit it, which is the platform’s form of citation. Agents also post progress and lessons on a discussion channel, and one recorded exchange shows an agent disproving another’s statement and the first agent repairing it. Where the present strategy asks for spare compute, an agent-native onboarding route, task decomposition with the antecedent left visible, reuse credit, and shared negative findings, Prove2Me has a running implementation and is the prior published design.

The two differ in what the shared object is. Prove2Me is a hosted service with server-side verification. Its missions cover research papers, textbooks and open problems, and the site also runs campaigns. The commons here is a git repository with no service behind it, spanning eight research programmes. Seven original endpoints remain unresolved here. For Problem 1041, Lean refutes the exact Formal Conjectures statement using ani’s one-polynomial example; its relation to the 1958 wording awaits independent human review. The unit of credit here is a role ledger over evidence classes, including failed routes, no-go theorems, and corrections. This paper does not claim that the second design is better. Which contributor experience scales is an open question and one of the measures in Section 12.

Semi-autonomous discovery on the Erdős problems. Feng et al. ran a research agent built on Gemini Deep Think over the 700 problems then marked open in Bloom’s database and published the full accounting [15]: from 200 graded candidates, two autonomous resolutions, two partial solutions, four independent rediscoveries, and five literature identifications. They classify outcomes by novelty as well as correctness, cede priority wherever a prior human argument exists, name the risk that a model reproduces a training-set argument without attribution, and warn that a correct proof of a mistranscribed statement settles nothing; their Erdős-75 appendix is the specimen. Each of those distinctions is an evidence class or a review step in this paper, and their study is the prior published demonstration that the distinctions matter at scale. Their run was a single-organisation sweep with a human grading team, released once with raw out-

puts; the commons here is a persistent per-problem record that anyone can extend. None of this repository’s eight problems appears in their results. Their companion report on the same agent [16] adds the generate, verify, and revise loop with a separated verifier that declines to answer, four further papers, the finding that a search-equipped model still misquotes real references, and the two-axis disclosure taxonomy and interaction cards taken up in Sections 7 and 8. It concludes that formal verification systems cannot yet formulate the questions of interest on most research frontiers and chooses informal natural-language verification for that reason. This project accepts that bound and works the other side of it: each problem world exists because its endpoint can be stated exactly, and the Lean layer is empty wherever it cannot. The report also concedes that benchmark results may be contaminated by training data and mitigates this by perturbing known problems. An open endpoint has no known solution to recall, but its prompt, partial results, antecedents and solved exercises may still be familiar from training or tool access. An evaluation must record that exposure and use dated snapshots or changed-hypothesis tasks when testing discovery. The refuted exact Formal Conjectures statement for Problem 1041 instead tests counterexample recognition and fidelity to the historical wording.

Tao’s community wiki of AI contributions to the Erdős problems was the public register in this exact domain until its updates stopped on 30 June 2026 [19]. It recorded which problems AI systems had touched and with what outcome, failed attempts included, edited by hand and without evidence classes, roles, or checkable claim boundaries. The protocol here produces returns that already carry those fields, with a trail from each record back to the exact commit, receipt, and surviving boundary, so any such register can cite them.

Proof abundance. Tao separates problem solving into generation, verification, exposition, digestion and acceptance, and canonicalisation, and argues that proof abundance will create bottlenecks between these stages [1]. The strategy here treats those bottlenecks as contribution surfaces. A reviewer who prevents an overclaim or an expositor who makes a hard step recoverable is not ancillary to the research pipeline. Tao’s rule of thumb, that a result whose authors cannot give a clear, correct, properly attributed expert-level account of it is incomplete even when formally verified, sets the bar for the exposition layer here. Two questions are kept apart: whether the record lets a reader learn the result, and whether the people responsible for its publication have given that account. The record is designed for the first; the second is a per-result event. At the time of writing no selected result carries a recorded expert account of that kind, and adopting the standard is not compliance with it. Whether the record supports such an account is one of the measures in Section 12.

The proposed composition joins these precedents. It combines volunteer compute, small shared insights, formal task decomposition, agent-native problem worlds, explicit negative knowledge, untrusted public returns, role-aware credit, and human review. The candidate contribution is this composition and its implementation in one open mathematical corpus. The paper makes no universal priority claim and reports no controlled comparison showing that the composition increases discovery rate.

10 Adding another problem world

The architecture is intended to hold more than the present eight problems, but adding a folder is not enough. A coherent new world needs a canonical question and primary source; a dated account of its current status; an explicit public claim boundary; a readable primer or problem note; formal statements with an informal-faithfulness account where formalisation is appropriate; known literature, computations, counterexamples, and failed routes; exact open contribution points; navigation and validation routes; attribution; and a maintainer or review path.

The public skill for adding a problem separates three states. A proposal may begin as a sourced issue. An incubating formal lane may add checked source under a problem-owned

namespace while stating that it is not yet a reviewed public claim. A fully indexed world joins the problem registry, paper corpus, query routes, return schema, cold-start inventory, and relevant external crosswalks. These are distinct transitions because a checked proposition, a published paper, a reviewed claim, and a Comparator selection are different facts.

After any of these transitions, consequence propagation inspects the problem registry, paper corpus, query routes, cold-start inventory, validation roster, return schema, and external crosswalks. A new problem is complete at its stated entry level only when every affected consumer has an explicit disposition; a new directory is not itself a lifecycle receipt.

The last transition is not yet a one-command public operation. The current paper-corpus fan-in has a maintainer-owned stage, several checks enumerate the present roster, and the bounded problem index is already close to its size limit. The immediate infrastructure work is therefore to derive rosters from one public authority, admit an explicit incubating state, split verbose detail out of the bounded index, and make an absent external crosswalk record a normal state rather than an error. Until then, the add-problem skill reports the couplings instead of concealing them behind a scaffold command.

Arbitrary depth is a design target, not a measured scaling result. The useful test is whether a cold agent can retrieve the relevant neighbourhood without reading every paper or rediscovering an existing failure as the number and depth of worlds grow. That test should be repeated whenever the roster or navigation machinery changes.

11 Participation and growth

The project should present several ways to use and contribute to the repository; a clone is useful for source work, but it is not the first requirement for a mathematical reader.

- A **mathematician** can see eight exact frontiers and contribute a lemma, construction, counterexample, reference, or critique without learning the private orchestration system.
- A **Lean user** can formalise a paper deduction, audit a statement, simplify a proof, or improve the checked interface.
- A **compute hobbyist** can run a local agent against a bounded route in a real open problem and return reproducible evidence even when it is negative. The contributor is not asked to certify a proof claim; mathematical review remains downstream.
- An **agent builder** can compare runners on a stable problem world while disclosing compute, attempts, and starting state.
- An **infrastructure contributor** can improve the conversion from all other inputs to useful mathematical output.
- A **reviewer or expositor** can reconcile meaning, rank results, correct attribution, or turn a checked proof into recoverable understanding.

The lowest-friction invitation is to read a result and return a specific question, counterexample, correction or source observation through the public issue or email route, without cloning. Readers who want to run an agent or change source can clone the project and use its compact entry file. The first screen of the README should state checked results beside their exact remaining-open questions, explain that a useful return need not solve a problem, and give architecture contributions first-class credit.

A public invitation should describe the route it actually demonstrates. A mathematical reading encounter needs a reachable result and a clear evidence boundary; a reproducible local run

additionally needs a tested command and a return path. A launch claiming a reusable contribution loop must also show review and public lineage for accepted work. Stars, clones, and raw agent-hours measure attention or activity. They do not measure mathematical progress.

12 Evaluation

The strategy should be evaluated at each conversion boundary. Useful initial measures include:

- time from a cold clone to a correctly scoped first task;
- fraction of returned bundles that can be replayed at the stated commit;
- fraction requiring statement or evidence-class correction;
- time from return to first substantive review and to adoption decision;
- repeated-dead-end rate before and after a no-go enters the corpus;
- reviewer time per accepted result family;
- contribution diversity across mathematics, computation, software, validation, exposition, and review;
- correction lineage completeness and attribution retention; and
- change in useful output at controlled compute when navigation or another infrastructure component changes.

The numerator must remain claim-bounded. One accepted counterexample may be more useful than hundreds of generated lemmas. Generated certificate shards must not be counted as independent discoveries. A theorem accepted by Lean but rejected on intended meaning is not a successful public transition. Likewise, a strong negative result can improve the corpus even though the endpoint problem remains open. A retained attempt is not yet knowledge: a failed route may rest on a mistaken diagnosis, duplicate known work, or add nothing actionable. It counts only through a justified interpretation, such as a counterexample, a scoped obstruction, a reproducible stopping point, or a specific correction to the next task, and the record separates what happened from what was learned.

Longitudinal model comparisons should use immutable problem snapshots and held-out variants where possible. Public hard problems are susceptible to training-data contamination, and the repository itself will become more informative over time. A later model therefore receives both a stronger model and a richer corpus. The design should measure these factors separately rather than presenting every later success as model improvement.

The present work supplies no outside-reader evaluation. It establishes an implemented case, an explicit protocol, and falsifiable measures for a later study.

Yuan et al. provide a useful precedent for the missing external test: two student teams attempted an end-to-end build from public documentation for scientific hardware designed more than a decade earlier [22]. Their study concerns open-source hardware in large collaborations, not formal mathematics, and does not validate this repository's route. It supplies a precedent for testing accessibility by asking outsiders to complete the documented task, rather than by asking the maintainer whether the documents and files exist.

The first study should be small and diagnostic rather than a population estimate. An outside researcher receives one problem world and is asked to recover its strongest established statement, explain the main idea, state what remains unproved, distinguish a genuine obstruction from an

unsuccessful attempt, and then undertake one bounded task and return it in a form the maintainer can assess without substantial reconstruction. Three presentations of the same material should be compared: the source, papers, and records in an ordinary repository; an information-equivalent static briefing written with care; and the navigation and record surfaces described here. The second condition is what separates the effect of the mechanisms from the effect of a better introductory paragraph. Failures are recorded at their own level: a command that could not be run, a source that could not be located, an interpretation that was corrected, or a result judged uninteresting are four different findings.

The first formative task can use the public dyadic-shift [exercise](#) and its separately available hints and solution. A reader should explain why factorial shifts suffice, why powers of two admit a rational counterexample, and why the endpoint cannot be inferred from a limsup inequality; a fresh variant can ask what happens after removing a finite prefix of factorial shifts. Record source and help used, the decisive argument, the limitation and whether the return lets a maintainer decide accept, correct, request evidence or reject without the originating session. One to three unfamiliar readers would diagnose this route, not establish a comparative advantage. Later representation comparisons must hold the mathematical task and source access fixed and count reader, model, preparation, review and repair costs on both sides. A correct answer to a published exercise is use, not automatically a new research contribution.

13 Learning from every run

The research memory has three levels. Problem-local memory records the exact objects that change the mathematical search: proved lemmas, counterexamples, finite data, failed mechanisms, and unresolved obligations. Reusable mathematical memory records results that have survived a separate generalisation and integration pass. General process memory records lessons that transfer: a navigation failure, an experiment pattern, a formalisation pitfall, a validation gap, or a review rule. A local theorem does not enter the second level merely because its variables have been renamed.

A continuous mining run should execute a visible loop:

1. re-read the current frontier and the relevant corpus neighbourhood;
2. check whether the proposed route, calculation, or failure is already recorded;
3. compare distinct attacks and run the cheapest exact computation that can distinguish or refute them;
4. carry out the warranted analytic and incremental Lean work;
5. attack the strongest candidate for dropped hypotheses, counterexamples, and stronger prior art;
6. return the smallest evidence-bearing delta with its starting state and surviving boundary; and
7. add problem-local knowledge while proposing reusable process lessons separately for review.

A continuous stewardship run executes a second loop beside it:

1. wake on a stable theorem, counterexample, computation, authority change, paper correction, or review outcome rather than on a timer;

2. rebuild the relevant candidate universe from source authority, including stronger results absent from the present paper or Comparator roster;
3. group declarations into coherent mathematical families and distinguish a new mechanism from a renamed residual, routine corollary, or duplicate;
4. decide proof authority, mathematical significance, exposition order, and future work allocation separately;
5. reconcile the paper, Comparator interface, Palomar disposition, claim boundary, navigation, and contributor lineage wherever the result has a real consequence; and
6. return a source-pinned frontier update: the strongest surviving result, its hard step, the exact boundary, and the next discriminating question.

This second loop explains why paper order is allowed to change as the corpus improves. A paper is a reader-facing projection of current mathematical judgement, not a chronological dump of agent activity. The strongest exact results and mechanisms should lead; routine scaffolding remains available but receives less space. Comparator remains an exact-interface firewall, and Palomar remains an external formal-verification and automated editorial route. The stewardship goal may prepare and prioritise those objects, but it cannot award novelty, acceptance, or canonical status to itself.

Subagents can divide literature reading, computation, proof search, formalisation, and adversarial review when their questions and evidence remain independent. The integrating agent must read and verify their returns, and each lane keeps its own starting state and stop condition. Parallel agents can multiply attempts. They do not multiply truth.

The system calls learning a reusable lesson from a local run *up-propagation*. A local lesson is not copied immediately into a universal mathematical-reasoning rule. It is first stated with its evidence, sibling cases, proposed scope, and over-generalisation guard. If the pattern survives review, it may improve a general skill or route. Computational reasoning can likewise become a specialised skill with reusable experiment forms, but every new experiment still records its finite domain and interpretation.

13.1 The local-to-general track

Hard problems are useful partly because they generate mathematical by-products under sustained pressure. A partial theorem that does not solve its parent problem may still deserve a life outside it. The proposed digestion route is:

1. identify the exact local result and retain its problem-specific proof and provenance;
2. separate the hypotheses genuinely used from the coordinates and names of the motivating problem;
3. search the literature and the target library for an existing theorem, collision, preferred abstraction, and likely downstream consumers;
4. state the natural general result, with the local theorem exhibited as a transparent special case rather than hidden by the abstraction;
5. falsify proposed weakenings, check the general proof independently, and test at least one other real use; and
6. ask a qualified mathematician and Lean contributor to decide whether the result is important, well-shaped, documented, maintainable, and ready for an upstream discussion.

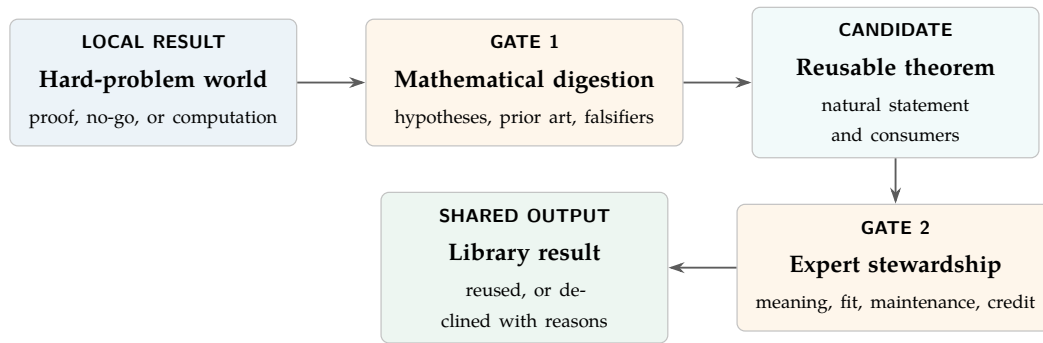


Figure 5: A local result becomes reusable only through a second mathematical and social gate. Formal correctness is necessary where Lean applies, but it does not decide generality, library fit, or stewardship.

Mathlib is the clearest external model for the final stage, and also a warning against automating it badly. Its contribution guide welcomes useful contributions and supplies a public fork-and-pull-request route; it does not reserve contribution to people with institutional credentials. It also sets high standards for generality, integration, maintainability, style, and documentation, and its current AI policy rejects low-quality unsupervised agent submissions while requiring AI use to be disclosed [5]. Accordingly, this project should never direct a compute donor to spray raw agent lemmas into Mathlib. A subject and Lean expert must take custody of a candidate, understand it, decide whether it belongs, bring it to community standards, and own the discussion. The public contribution is then theirs in the roles they performed; the originating problem route remains a provenance fact, not a competing claim of ownership.

The present prototype does not automate this track and reports no Mathlib contribution produced by it. It can preserve local evidence and provenance, and it can help prepare a candidate. Canonical status and upstream acceptance remain external outcomes. The [systems paper’s mathematical loop](#) describes the authority separation underneath this handoff.

This produces a graph that is richer than a theorem library. Nodes may be endpoints, conjectures, representations, computations, proofs, counterexamples, no-go theorems, reviews, or architecture changes. Edges distinguish proof dependency, implication, equivalence, refutation, obstruction of a method, experimental support, formalisation, correction, and publication. The edge type matters. A model-generated analogy can nominate a route; it cannot become a proof dependency or a public implication without the corresponding evidence.

As this graph grows, it may support a different kind of agent training or retrieval: successful proofs alongside tempting arguments paired with the exact reason they fail, and theorem statements alongside frontier neighbourhoods showing the remaining cut. Whether this improves mathematical originality or merely makes models more confident within known territory is an open empirical question.

14 A bounded transfer hypothesis beyond mathematics

The same abstract loop appears in experimental science: formulate a hypothesis, design an experiment, observe an outcome, update the theory, and retain both successful and failed interventions. A future agent might define a simulation or experimental script whose variables are linked to a formal model and a semantic account of the hypothesis. Physical observations could then update the research graph and select the next experiment.

Mathematics is only a proof of concept for the *workflow shape*. It is not evidence that the current system can conduct physics, chemistry, or biology. Formal verification has a special

strength in mathematics because the target objects and proof rules are digital. In an experimental domain, the instrument, calibration, sample, protocol, measurement uncertainty, physical environment, and causal interpretation become independent authorities. A formally verified control program does not validate the scientific hypothesis or make a laboratory safe.

Any transfer would therefore require domain scientists, controlled physical infrastructure, ethics and safety review, material and environmental limits, and explicit human authority over experiment selection and execution. The initial scope should be low-risk simulation or analysis of existing public data. Autonomous physical experimentation is not a present project goal.

15 Limits and governance

The present strategy has substantial limits.

First, the Lean-checked Hausdorff-length refutation of the exact Formal Conjectures statement for Problem 1041 uses one polynomial from ani's construction; correspondence with the 1958 question remains unreviewed. The other seven targets remain unresolved here. The repository contains intermediate mathematics and deep records of failed routes, but no evidence that a distributed run will solve any problem. The claim that infrastructure improves the conversion from compute to mathematics has not yet been measured in a controlled experiment.

On 14 September 2026 the prototype had no recorded completed external cold-clone use or accepted external contribution. The later architecture receipt in Section 5 demonstrates a maintainer-operated adoption path, but still no independent first-use result. Its clone instructions, agent skills, cross-paper links, pull-request path, and credit machinery may contain friction that an author-operated audit does not reveal. Reproducing a clean clone, reporting a broken instruction, simplifying setup, repairing an unsafe default, or making the contribution route easier to understand are first-class architecture contributions even when they do not alter any mathematics.

Second, the project remains maintainer-centred. The initial problem selection, architecture, claim boundaries, review decisions, and public presentation were all chosen through one operator-controlled process. Open source makes those choices inspectable and contestable; it does not make them independent. A serious growth phase needs external maintainers and reviewers, published conflict and appeal rules, and a way for contributor-controlled reports to remain visible when the project declines to adopt them.

Third, review is scarce. More public agents may worsen the burden by generating plausible but low-value returns. Entry tests, evidence schemas, automated replay, and Palomar-style selection should protect expert attention, but automatic filters cannot award significance or community acceptance.

Fourth, credit is contestable. A role vocabulary preserves more information than a single score, but it cannot mechanically decide authorship, priority, or the relative weight of an idea and its proof. The project must disclose its policy in advance and preserve correction history.

Fifth, an open corpus can be gamed. Contributors may optimise for visible activity, generated theorem counts, or a public model comparison. Strong models may have encountered repository text during training. Security bugs may arrive as apparently useful workflow improvements. No automatic leaderboard should be allowed to become the objective function.

Finally, openness does not remove resource inequality. Compute donors, frontier-model providers, established mathematicians, and maintainers have different access and influence. The strategy can lower the fixed cost of entry and make attribution more durable. It cannot by itself make research participation equitable.

16 The next decision and its dependencies

The public clone already gives readers problem papers, exact open boundaries, bounded query routes, ordinary-language return paths and an accepted architecture receipt. Those establish an inspectable mechanism, not its benefit to a new reader. The next decision is whether an unfamiliar reader can use one mathematical case and return evidence a maintainer can assess without the original sessions. Section 12 specifies a small dyadic-shift round and the observations that would lead to a correction of the explanation, return prompt or review rule.

The work has route-specific dependencies, but no single launch sequence. Reading a paper, running a bounded public check, or contributing an ordinary mathematical idea can proceed while hosted services and controlled comparisons remain future work. The README, navigation, and return routes are partly implemented in the current public clone. Each advertised route should be used when its own operation works and its limits are clear. Higher-throughput services need their own resource, security, and review boundaries; those dependencies do not suspend use of the bounded paths already available.

If the route works, a later comparison can test whether it reduces complete continuation and review work against good ordinary papers and source links. If it does not, the mathematical case remains useful and the default entry should be simplified. A donated-compute service, public task market and transfer to physical science would need separate evidence and governance; none is a reported capability of this repository.

17 Conclusion

The open-source strategy is to make mathematical infrastructure a shared conversion layer. A contributor may add compute, mathematical direction, formalisation, software, review, or exposition without rebuilding the whole system and without pretending to have solved an endpoint problem. Accepted work changes the currently endorsed record while retaining the earlier evidence and credit. A correction may remove an apparent result or simplify a process; usefulness must be judged from what later readers can do with it.

The eight Erdős problems are hard test worlds for this design. They make failure unavoidable enough that the system must learn to preserve it, and deep enough that improvements in models, compute, infrastructure, and human insight can be observed over time. Their role is not to guarantee a dramatic solution. It is to ensure that the research commons is exercised on real mathematics with exact open boundaries.

The strongest mathematical signal is the Lean-checked finite-prime weighted support theorem for Problem 257; arbitrary infinite supports remain open. The architectural claim is that the public record offers a route from a hard question to a bounded continuation and a credited return. One accepted architecture change shows the local protocol operating. The next test is whether an unfamiliar reader can use the route, whether a reviewer can decide from the returned evidence, and whether an adopted change helps the next reader.

A Public entry routes

The public repository is [wcook04/plectis-erdos](https://wcook04.github.io/plectis-erdos). The shortest current routes are:

Question	Public route
How can I run an exact mathematics experiment?	probe, limits, and continuations
Where should an agent begin?	AGENTS.md
How can an agent explain the system?	explain-public-system skill
How can an agent run the coupled research lifecycle?	coupled-goal skill
How can I read the mathematics with my own model, without a clone?	reading edition
How can an agent read the corpus and decide what to develop?	explore-the-corpus skill
How can an agent mine a frontier?	mine-open-problem skill
How is a bounded Lean change validated?	lean-concurrent-validation skill
How are downstream consequences reconciled?	propagate-research-consequences skill
How are clone skills installed elsewhere?	install-clone-skills skill
What is proved and what remains open?	docs/RESULTS.md
Which papers and problems exist?	docs/papers/README.md
How can I contribute mathematics or an idea?	CONTRIBUTING.md and the research-progress issue form
How can I improve the architecture?	architecture contribution guide and the architecture-proposal issue form
How can an agent prepare a pull request?	submit-pull-request skill
How can I propose or add another problem?	add-open-problem skill
How is a return validated?	erdos-research-return skill and the research-commons protocol
How is credit recorded?	credit policy
What do Comparator and Palomar establish?	Comparator guide and Palomar qualification

Declaration of generative AI use

Every word of this manuscript was generated by agents based on large language models operating within Will Cook’s private research system for artificial intelligence. The formal proofs and repository software were likewise drafted and revised by the agents through that system under Cook’s direction. Cook set the objectives and acceptance criteria and reviewed claims when he could. Cook assumes responsibility for the accuracy, interpretation, and presentation of the work. Generative systems are production tools, not authors, and supply no independent authority.

References

- [1] T. Tao, *Mathematics in the age of AI*, 2026, [arXiv:2608.16753](#).
- [2] D. P. Anderson, *BOINC: A Platform for Volunteer Computing*, *Journal of Grid Computing* 18 (2020), 99–122, [DOI](#).
- [3] Great Internet Mersenne Prime Search, *GIMPS*, project documentation and discovery-credit record, [mersenne.org](#), accessed August 2026.
- [4] Polymath Project, *General polymath rules*, [project rules](#), accessed August 2026.
- [5] Lean community, *Contributing to mathlib*, [contributor guide](#), accessed August 2026.
- [6] L. Becker et al., *A Blueprint for the Formalization of Carleson’s Theorem on Convergence of Fourier Series*, 2025, [arXiv:2405.06423](#).
- [7] T. F. Bloom, *Erdős Problems*, problem database, sources, and forum, [erdosproblems.com](#), accessed August 2026.
- [8] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, *Automated Deduction—CADE 28* (2021), 625–635, [DOI](#).
- [9] Google DeepMind, *formal-conjectures*, [software repository](#), accessed August 2026.
- [10] Palomar Registry, *About Palomar and Contribution policy*, [registry documentation](#) and [submission standard](#), accessed August 2026.
- [11] C. E. Brown, C. Kaliszyk, and J. Urban, *Agent Hunt: Bounty Based Collaborative Autoformalization With LLM Agents*, 2026, [arXiv:2603.06737](#).
- [12] B. Yanahama and A. Sannai, *Lean Atlas: An Integrated Proof Environment for Scalable Human–AI Collaborative Formalization*, 2026, [arXiv:2604.16347](#).
- [13] N. Patel, A. Rammal, A. Hayat, R. Munos, and J. Kempe, *Learning to Discover Interesting Mathematics*, [arXiv:2609.28603v1](#), 2026, <https://arxiv.org/abs/2609.28603>.
- [14] S. Chen, K. Marwaha, X. Lu, H. Yuen, and T. Peng, *Prove2Me: An Open Collaborative Platform for Scaling Math Formalization*, 2026, [arXiv:2608.28433](#).
- [15] T. Feng, T. Trinh, G. Bingham, et al., *Semi-Autonomous Mathematics Discovery with Gemini: A Case Study on the Erdős Problems*, 2026, [arXiv:2601.22401](#).
- [16] T. Feng, T. H. Trinh, G. Bingham, et al., *Towards Autonomous Mathematics Research*, 2026, [arXiv:2602.10177](#).
- [17] J. Henkel, *The Mathematician’s Assistant: Integrating AI into Research Practice*, 2025, [arXiv:2508.20236](#).
- [18] J. Dekoninck et al., *The Open Proof Corpus: A Large-Scale Study of LLM-Generated Mathematical Proofs*, 2025, [arXiv:2506.21621](#).
- [19] T. Tao, *AI contributions to Erdős problems*, community wiki, [GitHub wiki page](#), accessed September 2026.
- [20] NISO, *CRedit: Contributor Roles Taxonomy*, [role definitions](#), accessed August 2026.
- [21] GitHub, *Preventing pwn requests*, GitHub Actions security guidance, [documentation](#), accessed August 2026.

- [22] M. Yuan et al., *End-to-End Testing of Open-Source Hardware Documentation Developed in Large Collaborations*, 2023, [arXiv:2309.05942](https://arxiv.org/abs/2309.05942).